What is a number?

- Represent a value / quantity

How to write a number?

Base - 10 → 10 digits

0 1 2 3 4 5 6 7 8 9

$$\underset{\text{hundreds}}{\frac{6}{10^2}} \quad \underset{\text{tens}}{\frac{2}{10^1}} \quad \underset{\text{ones}}{\frac{5}{10^0}}$$

Binary numbers

Base 2

only can use 2 digits, 0 & 1

1	1
2^1	2^0

$1 \times 2^1 \quad 1 \times 2^0$

$1 \times 2 \quad 1 \times 1$

$2 + 1 = 3$

$(11)_2 = (3)_{10}$

$(1111)_{10}$

"one thousand one hundred and eleven"

$(1111)_2 = (15)_{10}$

* The same string of digits can have different numerical values depending on which base is being used!

$\frac{1}{2^3}\ \frac{1}{2^2}\ \frac{1}{2^1}\ \frac{1}{2^0}$

$2^3 + 2^2 + 2^1 + 2^0$

$8 + 4 + 2 + 1$

$(15)_{10}$

① $1011 \rightarrow \frac{1}{2^3}\ \frac{0}{2^2}\ \frac{1}{2^1}\ \frac{1}{2^0} = (11)_{10}$

② $01111 \rightarrow \frac{0}{2^4}\ \frac{1}{2^3}\ \frac{1}{2^2}\ \frac{1}{2^1}\ \frac{1}{2^0} = (15)_{10}$

③ $10001 \rightarrow \frac{1}{2^4}\ \frac{0}{2^3}\ \frac{0}{2^2}\ \frac{0}{2^1}\ \frac{1}{2^0} = (17)_{10}$

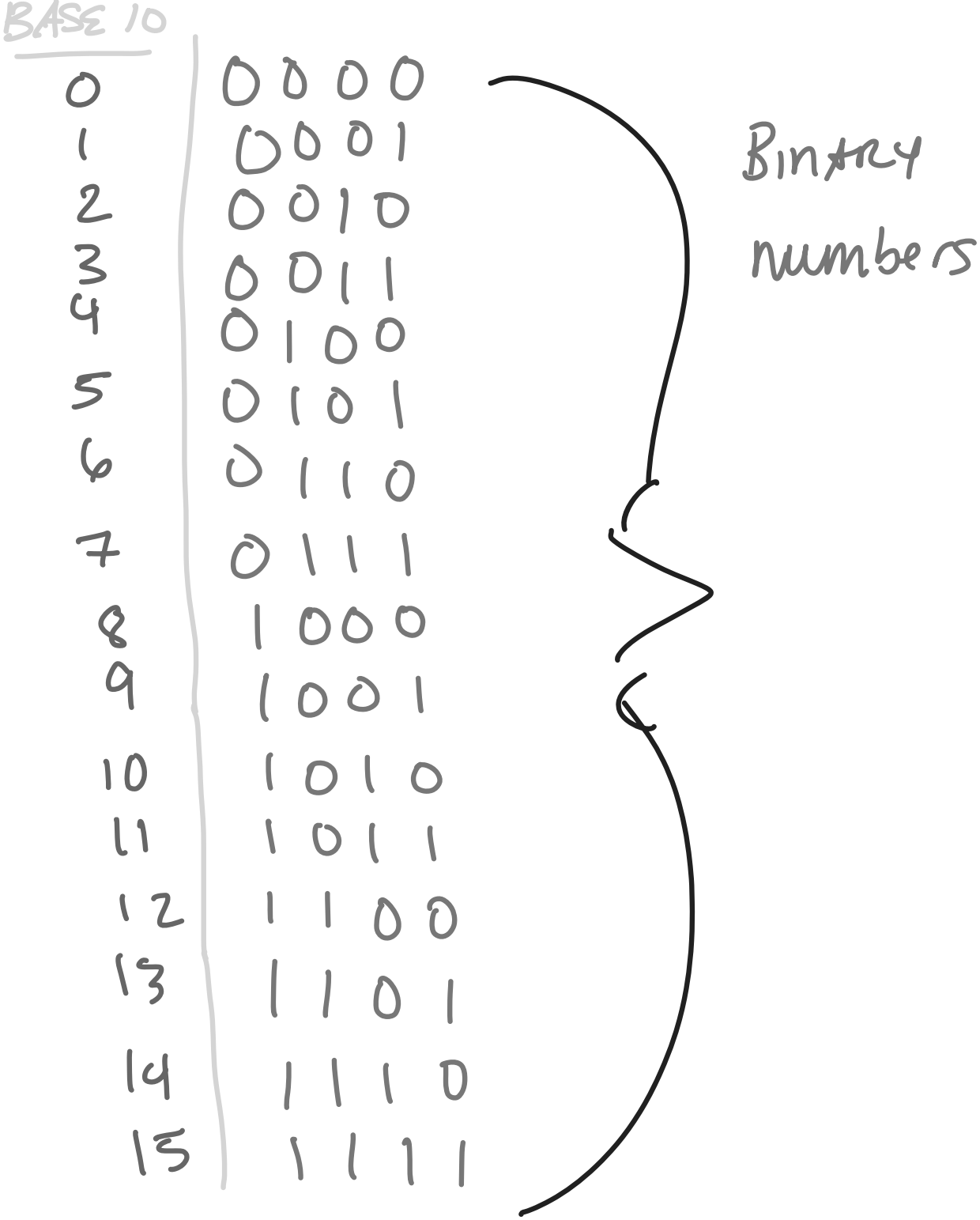
④ $10101 \rightarrow \frac{1}{2^4}\ \frac{0}{2^3}\ \frac{1}{2^2}\ \frac{0}{2^1}\ \frac{1}{2^0} = (21)_{10}$

The binary numbers are converted to base 10 by adding the place values where there is a 1

es $(1011)_2 = (11)_{10}$ $\frac{1}{2^3}\ \frac{0}{2^2}\ \frac{1}{2^1}\ \frac{1}{2^0}$

Because $8 + 2 + 1 = 11$

2^3 2^1 2^0



★ Practice counting from 0 - 15 in binary! ★

Numbers for quantity/amount/size

unsigned number

Signed numbers (+/-)

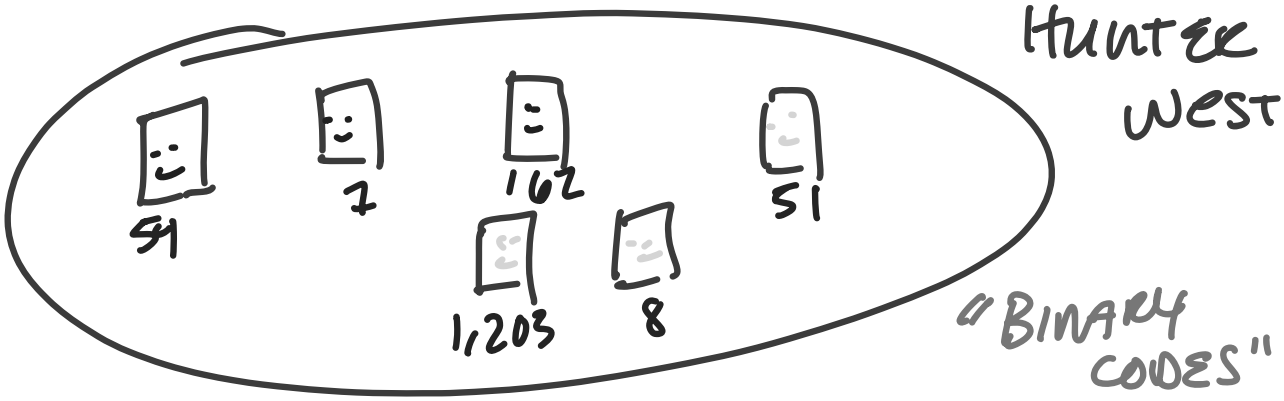
signed-magnitude

$+(101)_2$ $+(7)_{10}$ $-(352)_{10}$

$-(100)_2$

Complement of a number also represent positive + negative

Assigning unique numbers to items in a set.



Quiz 1 powers of 2 up to 2^{10}

Table 1.1
Powers of Two

n	2^n	n	2^n
0	1	8	256
1	2	9	512
2	4	10	1,024 (1K)
3	8	11	2,048
4	16	12	4,096 (4K)
5	32	13	8,192
6	64	14	16,384
7	128	15	32,768

$2^5 + 2^5 = 2^6$

$32 + 32 = 64$

$2^6 + 2^6 = 2^7$

$64 + 64 = 128$

Convert Decimal 42 and 31 to Binary

Section 1.3 ex) 1.1

★ First remainder is LSB ★

			R	2^n
42/2	=	21	= 0	2^0
21/2	=	10	= 1	2^1
10/2	=	5	= 0	2^2
5/2	=	2	= 1	2^3
2/2	=	1	= 0	2^4
1/2	=	0	= 1	2^5

MSB

$(101010)_2 = (42)_{10}$

Unsigned binary #; all bits contribute to value

		Rem
31 / 2 =	15	1
15 / 2 =	7	1
7 / 2 =	3	1
3 / 2 =	1	1
1 / 2 =	0	1

STOP when whole num. portion = 0

$(11111)_2 = (31)_{10}$

Radix complement

BINARY → 2's complement

Diminished Radix complement

(R-1)'s comp

BINARY → 1's complement

Radix complement =

diminished radix comp + 1

2's complement =

1's complement + 1

To get 1's comp. you only need to flip the bits

1010 → 0101

This is the one's comp of 1010

$(10)_2 = (2)_{10}$

```
  0 0 1
  0 1 0 1
+       1
-------------
  0 1 1 0
```

0110 → 2's complement of 1010

Find the 1's comp and the 2's comp of the following

① $101010 \rightarrow$ 010101 1's comp, 010110 2's comp

② $1100111 \rightarrow$ 0011000 1's comp, 0011001 2's comp

$$\begin{array}{r} {\scriptstyle 1} \\ 010101 \\ +\quad\ 1 \\ \hline 010110 \end{array} \qquad \begin{array}{r} 0011000 \\ +\quad\ 1 \\ \hline 0011001 \end{array}$$

★ Section 1.6 ★

Table 1.3
Signed Binary Numbers

Decimal	Signed-2's Complement	Signed-1's Complement	Signed Magnitude
+7	0111	0111	0111
+6	0110	0110	0110
+5	0101	0101	0101
+4	0100	0100	0100
+3	0011	0011	0011
+2	0010	0010	0010
+1	0001	0001	0001
+0	0000	0000	0000
−0	—	1111	1000
−1	1111	1110	1001
−2	1110	1101	1010
−3	1101	1100	1011
−4	1100	1011	1100
−5	1011	1010	1101
−6	1010	1001	1110
−7	1001	1000	1111
−8	1000	—	—

$+(111)_2$
↓
0

Sign bit is always MSB

1
↑
$-(111)_2$

$+(11)_2$ $(-3)_{10}$

011

ONE'S COMP

signed 1's comp. Representation

100

signed mag

0 0 1 1
2^2 2^1 2^0

Sign bit

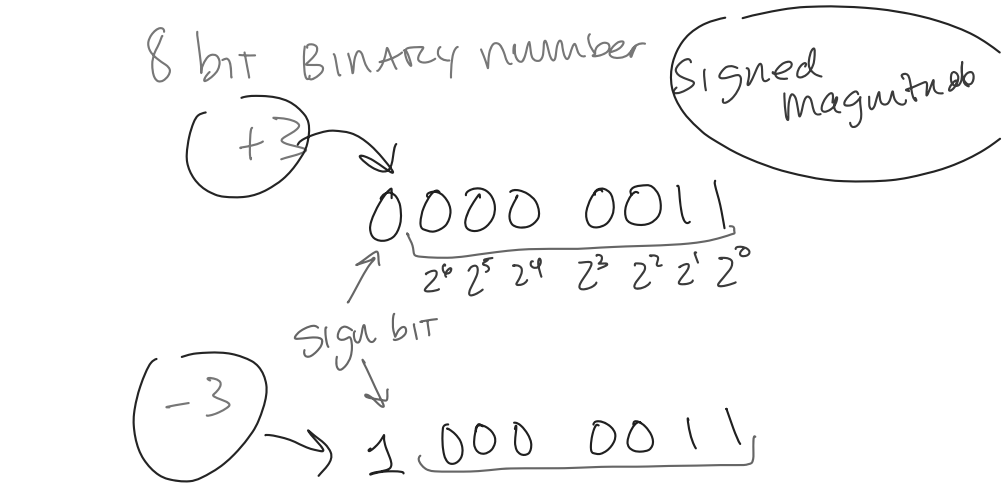
value BITS

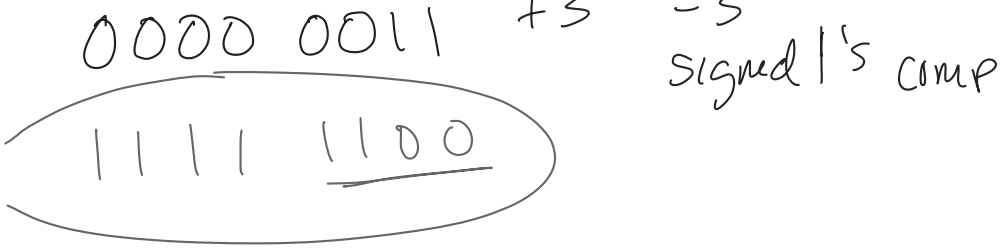
Section 1.7

$$(185)_{10} = (0001\ 1000\ 0101)_{BCD} = (10111001)_2$$

Table 1.4
Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001





WRITE Integers -1 to -7

in signed one's complement

all numbers should have 4 BITS

① get positive Rep. Make sure it has 4 BITS

② get 1's comp of positive This is your negative #

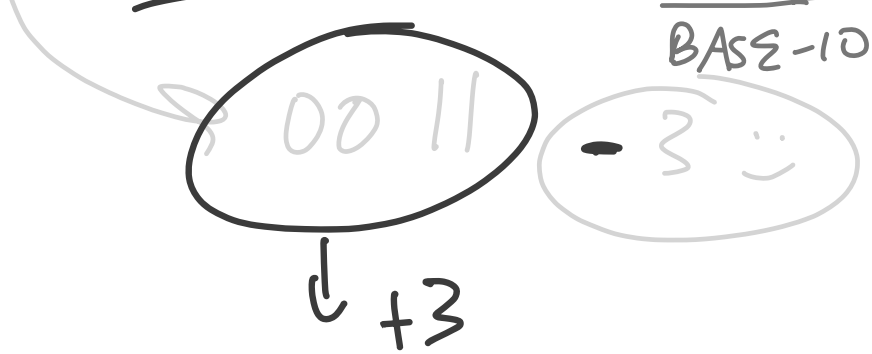
③ Take one's comp of that # and you go back to positive

Signed 1's comp binary number

1100 = -3

How do we write it as a signed magnitude decimal #

BASE-10



+5

0101

−5 signed 1's comp

1010

0101

$\bar{2}^3$ $\bar{2}^1$ $\bar{2}^0$

How do we write it as a signed magnitude decimal #

−5

Signed 2's comp #'s

(one's comp) +1 = 2's comp

How do we write them as a signed magnitude decimal #

① 1111 → 0000 +1 = 0001 (−1)

② 1101 → 0010 +1 = 0011 (−3)

③ 0011 (+3)

④ 0111 (+7)

⑤ 1001 → 0110 + 1 = 0111 (−7)